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The Role of AI in ASEAN Social Impact

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1.0 Introduction

Flood is one of the most frequent and destructive natural disasters worldwide, particularly in Asia where seasonal monsoons, heavy rainfall, and rapid urbanization significantly increase flood risks. According to the United Nations Office for Disaster Risk Reduction (UNDRR), floods account for nearly 44% of all disaster events globally, affecting billions of people over the past decades. Asia is considered the most flood-prone region, with countries such as Malaysia, Indonesia, Bangladesh, Thailand, India, and China experiencing recurring floods every year due to monsoon systems and tropical storms.

In Southeast Asia specifically, the annual monsoon season brings intense rainfall that frequently leads to severe flooding. The Asian Development Bank reports that over 1.8 billion people in Asia are exposed to flood risks, with urban populations being particularly vulnerable due to high population density and limited drainage infrastructure. Flood disasters not only threaten human lives but also cause extensive damage to homes, infrastructure, agriculture, and transportation networks.

But first of all, let's look locally. Malaysia is one of the countries significantly affected by seasonal flooding. Data from the Department of Irrigation and Drainage Malaysia (Jabatan Pengairan dan Saliran, JPS) shows that floods occur almost every year during the Northeast Monsoon (November to March). During this period, heavy rainfall often results in rising river levels and flash floods across several states, including Kelantan, Terengganu, Pahang, Johor, and parts of Selangor. One notable example is the 2021–2022 Malaysian floods, which displaced more than 125,000 residents and caused billions of ringgit in damages. This disaster highlighted the urgent need for improved flood preparedness and early warning systems.

Flood safety plays a crucial role in reducing casualties and minimizing economic losses. Early warning systems, timely evacuation guidance, and accurate real-time information can significantly improve community preparedness and response. However, many communities still face challenges in accessing reliable and real-time flood updates, evacuation routes, and safety instructions during emergencies.

To address these challenges, this project proposes MyBanjir, a mobile-based flood early warning and information application designed to enhance disaster preparedness and community safety. The application aims to provide users with real-time flood alerts, safety guidelines, and accessible disaster information through an intuitive mobile platform. By leveraging mobile technology, MyBanjir seeks to bridge the information gap between authorities and the public, enabling faster decision-making during flood emergencies.

Through this initiative, MyBanjir contributes to improving disaster resilience in Asia by promoting awareness, preparedness, and proactive flood risk management. The development of such digital tools aligns with global disaster reduction initiatives such as the Sendai Framework for Disaster Risk Reduction, which emphasizes the use of technology and information systems to strengthen early warning capabilities and reduce disaster impacts on communities.

2.0 Review on Existing Solutions

Flood early warning systems are important to help communities prepare and respond quickly during flood events. Several mobile applications already provide flood alerts and river monitoring features. However, many of these solutions mainly focus on basic notifications and lack advanced capabilities such as predictive analysis, community reporting, and smart evacuation guidance. Therefore, this comparison highlights the strengths and limitations of existing solutions and shows how the proposed system can improve current approaches.

Feature\Product	MyBanjir (Us)	MyPublicInfoBanjir	FloodAlert	Sachet
Flood alerts	Yes	Yes	Yes	Yes
Real-time river monitoring	Yes	Yes	Yes	Limited
Predictive flood model	Yes	Limited	No	Limited
Hazard map	Yes	Yes	No	Yes
Community reporting	Yes	No	No	Partial
Evacuation routing	Yes	No	No	Partial
AI / analytics	Yes	No	No	No
IoT integration	Yes	Partial	No	No

From the comparison, most existing applications provide basic flood alerts and monitoring. However, they lack integrated features such as AI-based prediction, community reporting, and IoT sensor support. The proposed system aims to combine these features into one platform to provide more accurate information and better support for flood preparedness and response.

2.1 Market Opportunity

Flooding is becoming more frequent due to climate change and urban development. This increases the demand for smarter early warning systems that can provide faster and more reliable information. A solution like **MyBanjir** can support communities, local authorities, and disaster management agencies by offering real-time monitoring, predictive insights, and improved decision-making tools.

2.2 Investor Value

This platform has strong potential for collaboration with government agencies, municipalities, and smart city initiatives. Investors can benefit through partnerships, system deployment, and data services for disaster management. At the same time, the system contributes to improving public safety and reducing flood-related risks.

3.0 Methodology

This part describes the methodology used in the development of the MyBanjir system. The methodology outlines the overall approach taken to design, develop, and implement the proposed flood management application. It explains the processes involved in identifying the problems related to existing flood monitoring systems, defining system requirements, designing the system architecture, and integrating intelligent technologies to enhance system capabilities.

The purpose of this methodology is to ensure that the system is developed in a structured and systematic manner so that it can effectively address real-world flood management challenges. In addition, this methodology also describes the integration of Artificial Intelligence (AI) within the system to support predictive analysis, safety recommendations, and user assistance during flood situations.

The methodology is organized into several sections including the development approach, problem identification, system architecture and functional modules, and AI integration within the system.

3.1 Development Approach

The development of the MyBanjir system follows a structured system development approach that focuses on solving real flood-related problems through the integration of real-time monitoring, emergency response tools, and intelligent analysis.

The development process consists of several stages including problem identification, requirement analysis, system development, and Artificial Intelligence integration. These stages ensure that the system is able to provide accurate information and practical support to users during flood situations.

The approach emphasizes building a centralized platform that allows users to monitor flood risks, receive emergency alerts, locate nearby shelters, and access safety information

quickly. By combining multiple features into a single platform, the system aims to improve community preparedness and response during flood disasters.

3.2 Problem Identification

Based on the review of existing flood monitoring applications, several limitations were identified. Most existing solutions mainly provide basic flood alerts and river monitoring but do not offer advanced support features that help users make better decisions during emergencies.

During flood situations, users often face difficulties in obtaining accurate and timely information. For example, individuals may not know the safest evacuation routes, the nearest available shelters, or the severity of the flood risk in their location. In addition, many applications do not provide predictive insights that can help users prepare before flooding occurs.

These limitations highlight the need for a more intelligent and integrated flood management system. Therefore, the MyBanjir application is developed to address these issues by combining real-time monitoring, predictive analysis, emergency assistance, and safety guidance within a single platform.

3.3 System Architecture and Functional Modules

The MyBanjir system is developed with several functional modules that support flood monitoring, emergency response, and disaster preparedness.

3.3.1 Weather Monitoring

The weather monitoring module provides users with real-time weather information such as temperature, rainfall conditions, and a seven-day forecast. The system also includes a precipitation radar map that allows users to observe rainfall distribution and identify potential storm patterns that may lead to flooding.

In addition, the system provides a flood forecast section that shows predicted flood probability and estimated water levels for the upcoming days. This helps users understand potential risks and prepare in advance.

3.3.2 Flood Alert and Notification

The notification module provides important updates related to environmental conditions and disaster risks.

The system generates several types of alerts, including:

- Rain alerts when heavy rainfall is expected
- River level warnings when water levels approach dangerous thresholds
- Shelter availability updates
- Road closure notifications caused by flooding

These alerts help users stay informed about the current situation and allow them to take early precautionary actions.

3.3.3 Shelter Locator and Evacuation Routing

During flood emergencies, locating a safe evacuation center is critical. The Nearby Shelter module allows users to identify evacuation centers located near their current location.

The system displays available shelters together with estimated travel time and distance. This allows users to determine the fastest way to reach a safe location during an emergency.

3.3.4 Emergency SOS

The MyBanjir system includes an Emergency SOS feature that allows users to quickly request help when they are in danger.

When activated, users can send an emergency request together with their location and additional information describing their situation. The system confirms when the request has been successfully submitted so that users are aware that assistance is being dispatched.

This feature helps improve emergency response time and allows rescue teams to locate affected individuals more efficiently.

3.3.5 Safety and Preparedness

The system also provides a Safety module to help users prepare before, during, and after flood events.

This module includes educational resources such as emergency kit preparation, evacuation guidelines, and survival tips. A structured Safety Checklist is also provided to guide users through the different stages of a flood disaster.

The checklist is divided into three categories:

- **Before Flood** – preparation steps such as storing emergency supplies and securing important documents

- **During Flood** – safety actions such as moving to higher ground and avoiding flooded areas

- **After Flood** – recovery steps including checking for structural damage and avoiding contaminated water

This feature helps improve public awareness and encourages proactive disaster preparedness.

3.4 Artificial Intelligence Integration

Artificial Intelligence (AI) is integrated into the MyBanjir system to enhance flood awareness, decision-making, and emergency response capabilities. The AI components assist users by analyzing environmental data, providing safety recommendations, and supporting communication during emergency situations.

Several AI-driven functionalities are implemented within the system modules.

3.4.1 AI Weather Analysis

Within the weather monitoring module, AI is used to analyze weather conditions such as rainfall intensity, precipitation patterns, and forecast data. By processing this environmental data, the system can detect potential weather patterns that may lead to flooding.

The AI analysis helps users better understand possible flood risks by interpreting weather data and presenting it in a more meaningful way. This allows users to anticipate dangerous weather conditions and take precautionary actions earlier.

3.4.2 AI Safety Suggestions

The Safety module incorporates an AI-based recommendation feature that provides users with personalized safety guidance during flood situations.

Based on the user's current situation and environmental conditions, the AI system can suggest important safety actions. For example, the system may recommend preparing emergency kits, moving to higher ground, or avoiding certain areas that are likely to flood.

These AI-generated suggestions help users make faster decisions and improve their preparedness during disaster situations.

3.4.3 AI Chatbot Assistance

The MyBanjir system also includes an AI-powered chatbot that allows users to obtain quick assistance and information during emergencies.

Users can interact with the chatbot to ask questions related to flood safety, evacuation procedures, nearby shelters, and emergency preparation. The chatbot provides immediate responses and guidance, helping users access important information without needing to navigate through multiple pages.

This feature improves accessibility and ensures that users can receive assistance quickly during stressful situations.

3.4.4 AI Shelter Analysis

In the Nearby Shelter module, AI is used to analyze location data and identify the most suitable evacuation shelters for users.

The system evaluates factors such as distance, estimated travel time, and accessibility to recommend nearby shelters. By analyzing this information, the AI system helps users determine the safest and most efficient evacuation option during flood emergencies.

This intelligent routing assistance helps reduce confusion and improves evacuation efficiency when time is critical.

4.0 AI Acknowledgement

Artificial Intelligence (AI) tools were utilized during the development of the MyBanjir to support concept development, system design, and refinement of the written report. AI-assisted tools such as ChatGPT and Gemini were used to assist in brainstorming ideas, organizing technical explanations, improving language clarity, and structuring sections of the documentation.

In addition, AI technologies form a core component of the proposed system architecture. Machine learning and data analytics techniques are incorporated to support flood prediction, evacuation route optimization, and disaster response decision-making. These AI components analyze environmental datasets such as rainfall intensity, river water levels, terrain elevation, and historical flood patterns to generate predictive insights and early warning alerts.

The system also integrates intelligent features including safe evacuation route recommendations, community-based disaster reporting verification, and resource optimization for evacuation shelters. These AI-driven functionalities aim to improve situational awareness, enhance disaster preparedness, and support more efficient emergency response.

All AI-assisted outputs and system design suggestions were carefully reviewed, evaluated, and refined by the project team to ensure accuracy, relevance, and alignment with the project objectives. The final analysis, system concept, and conclusions presented in this report represent the original work and responsibility of the authors.

5.0 Output/UI screenshot

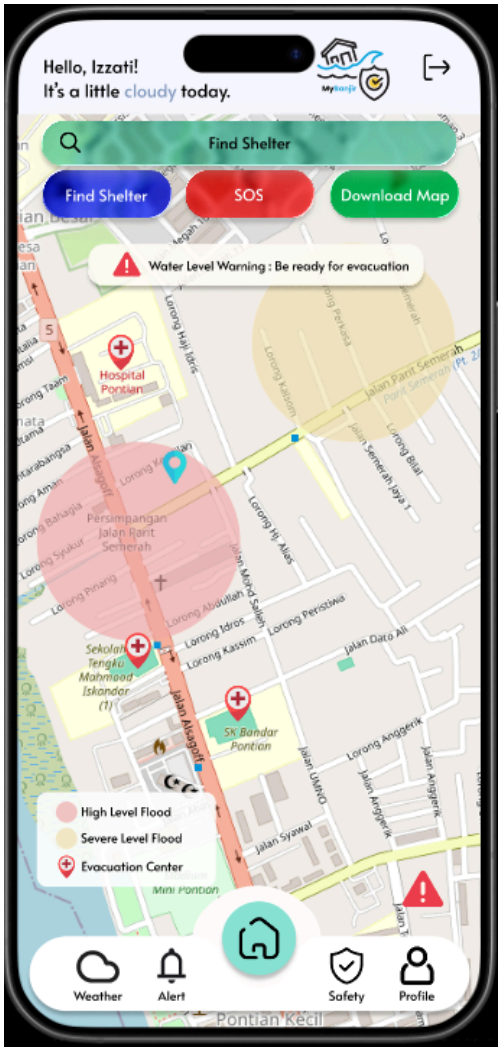


Figure 5.1 Home Page

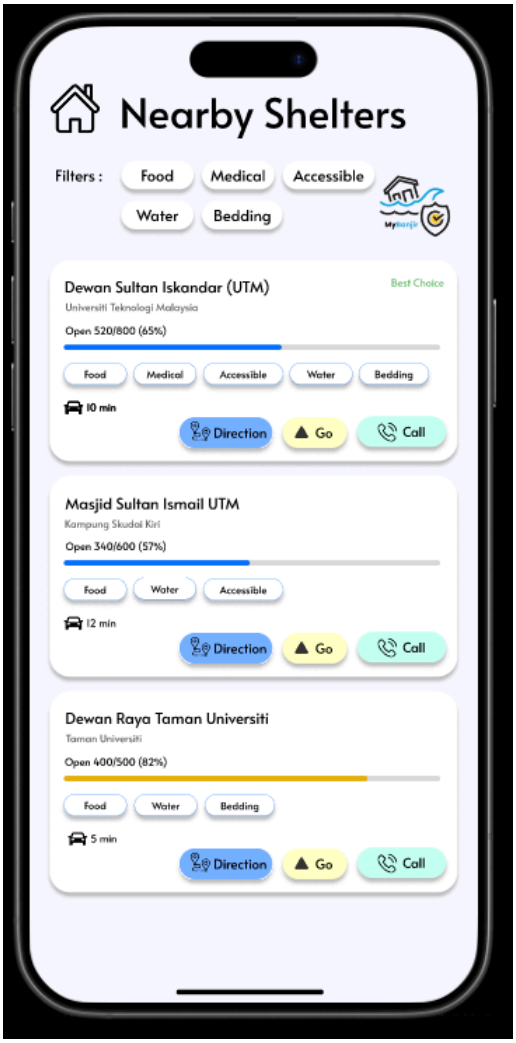


Figure 5.2 Shelters List Page

Figure 5.3 Shelters Navigation

Figure 5.4 Emergency SOS

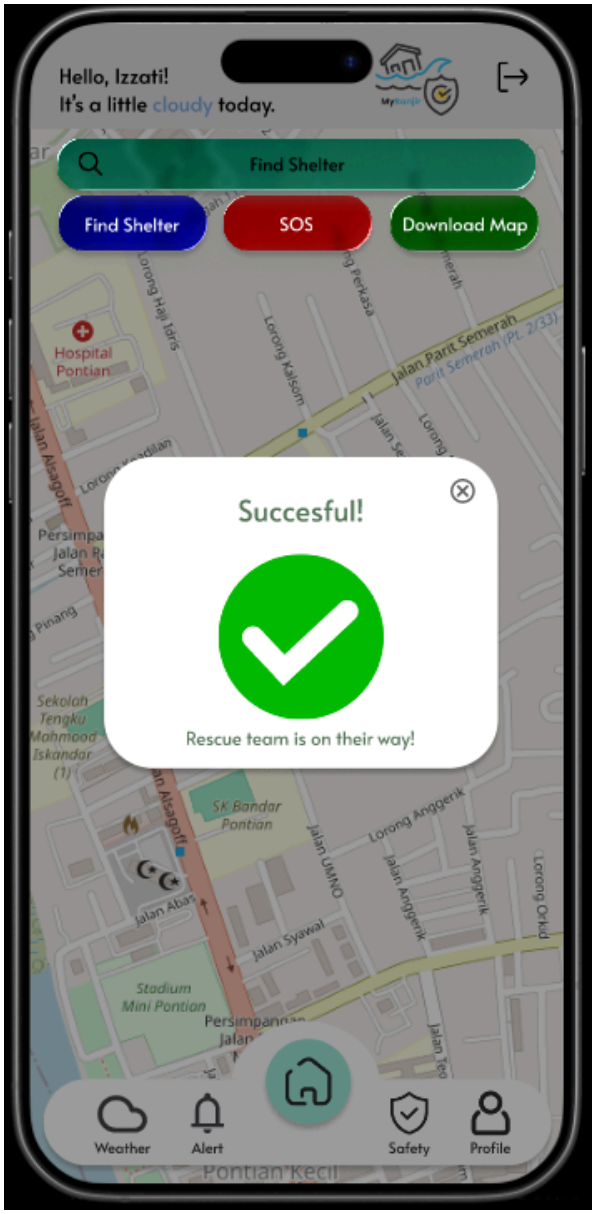


Figure 5.5 Successful Report SOS

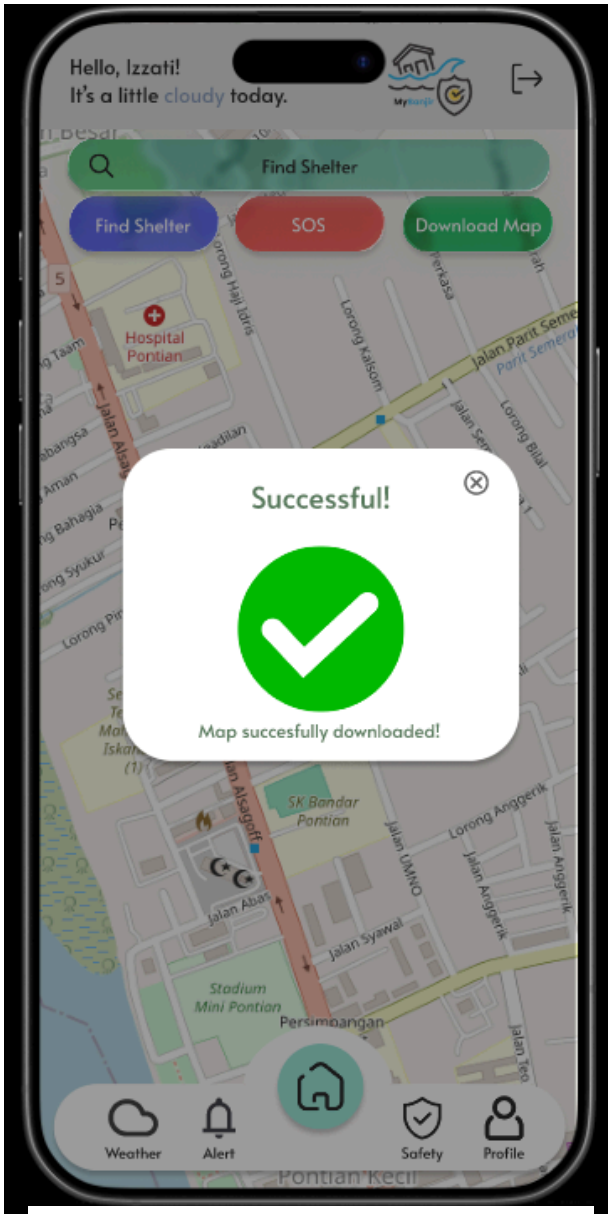


Figure 5.6 Successful download map

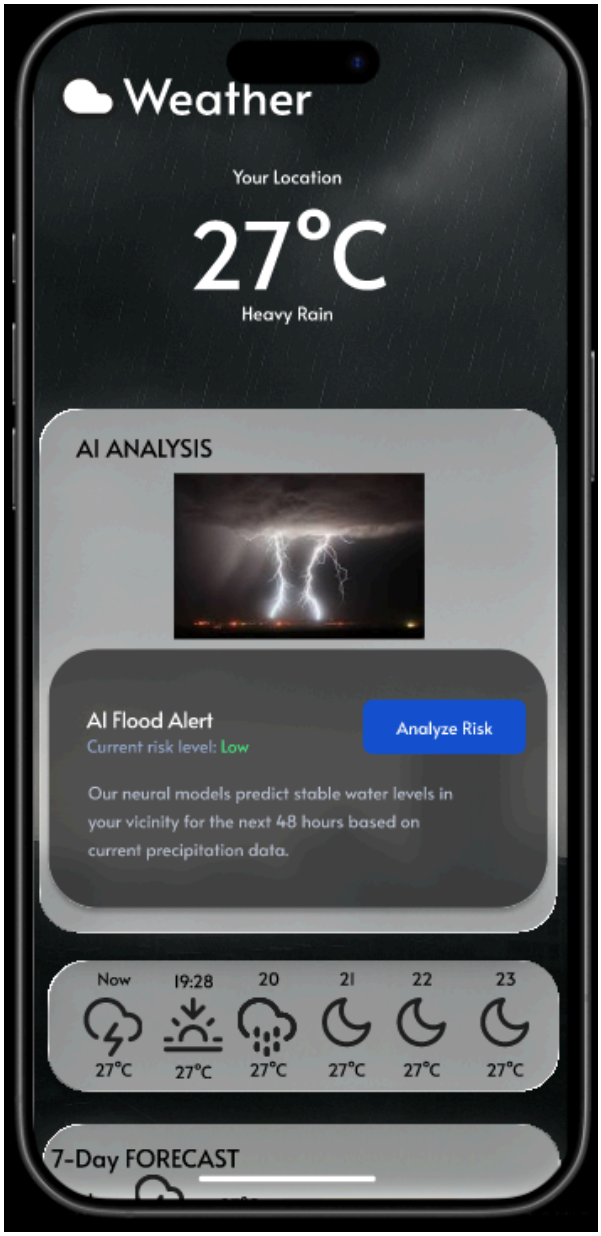


Figure 5.7 Weather Page

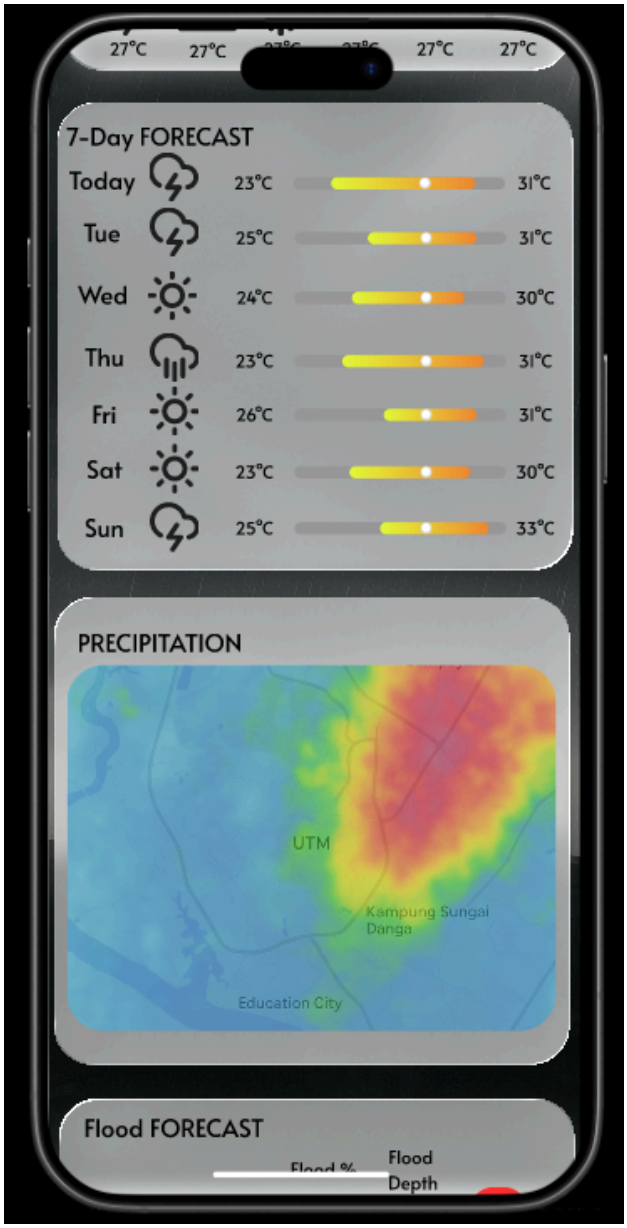


Figure 5.8 Weather Page

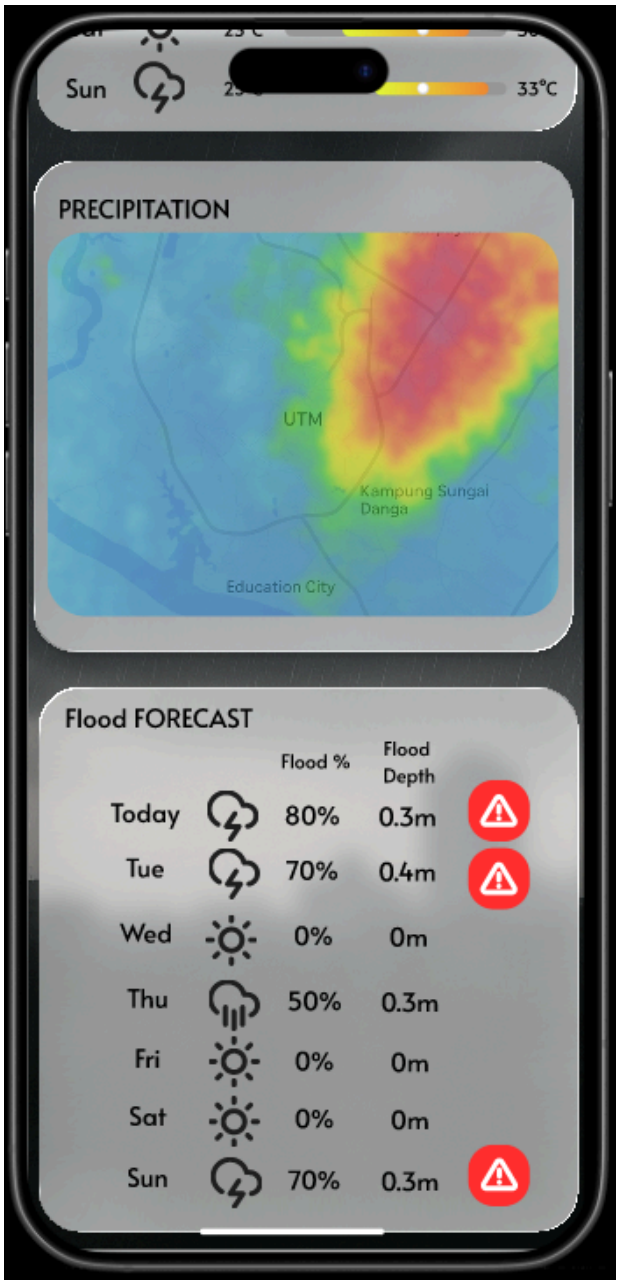


Figure 5.9 Weather Page

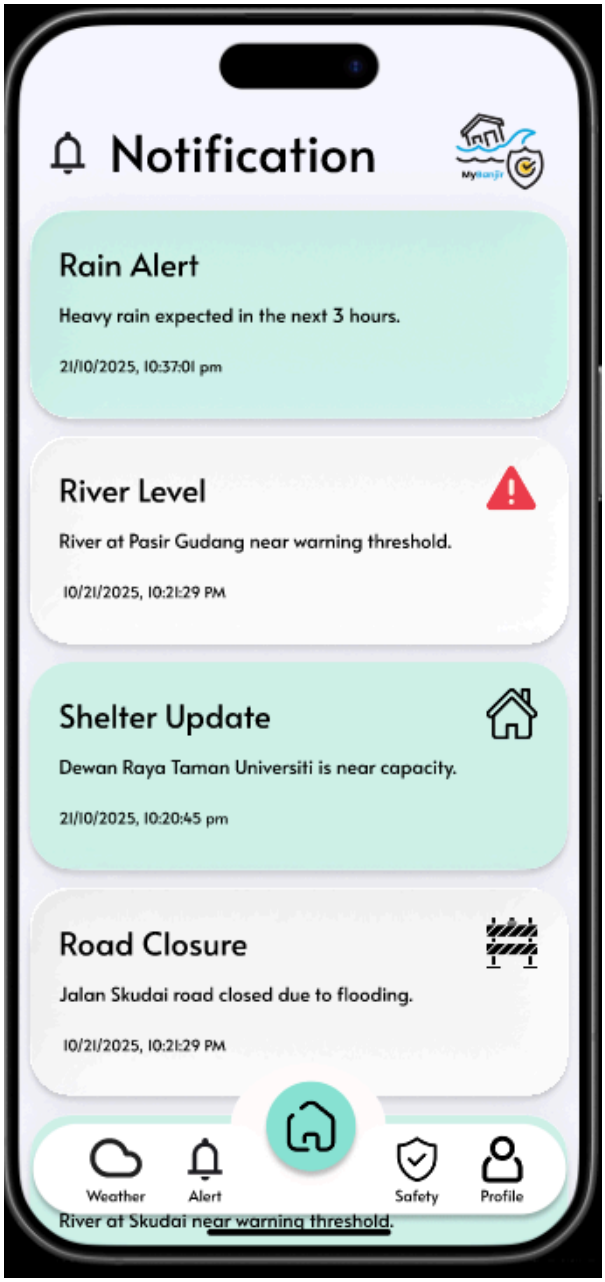


Figure 5.10 Alert Page

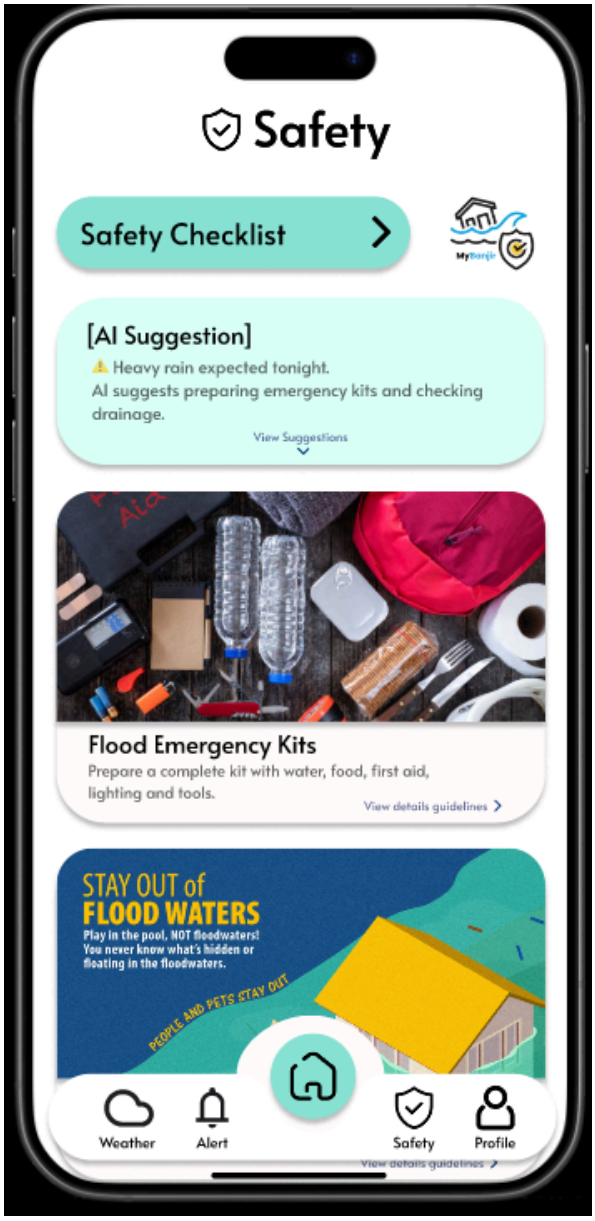


Figure 5.11 Safety Page

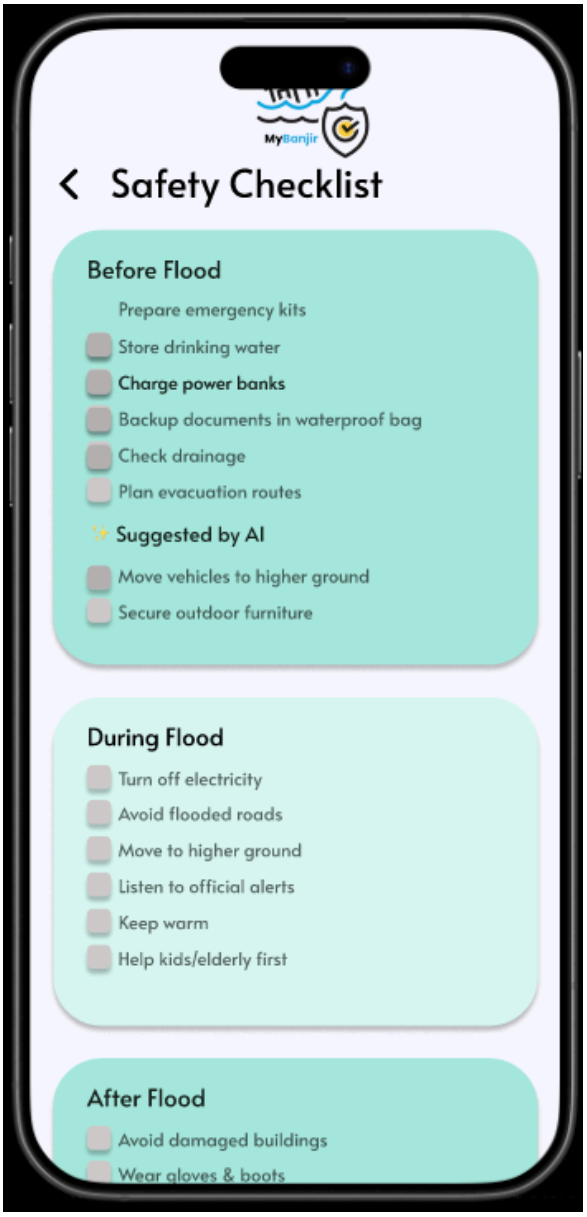


Figure 5.12 Safety Checklist Page

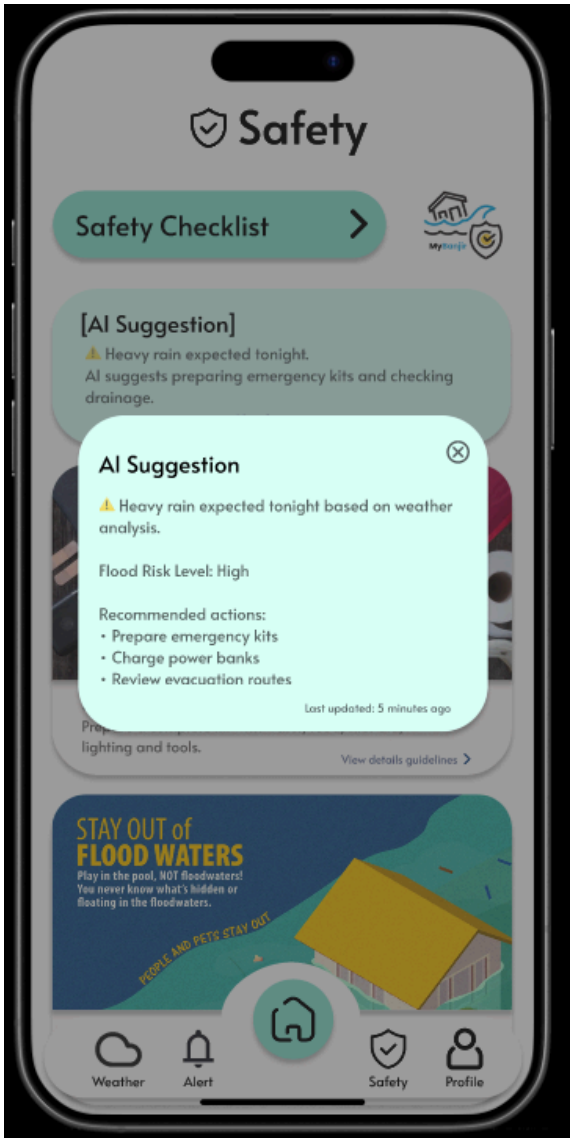


Figure 5.13 AI Safety Suggestion

Figure 5.14 Flood Emergency Kits

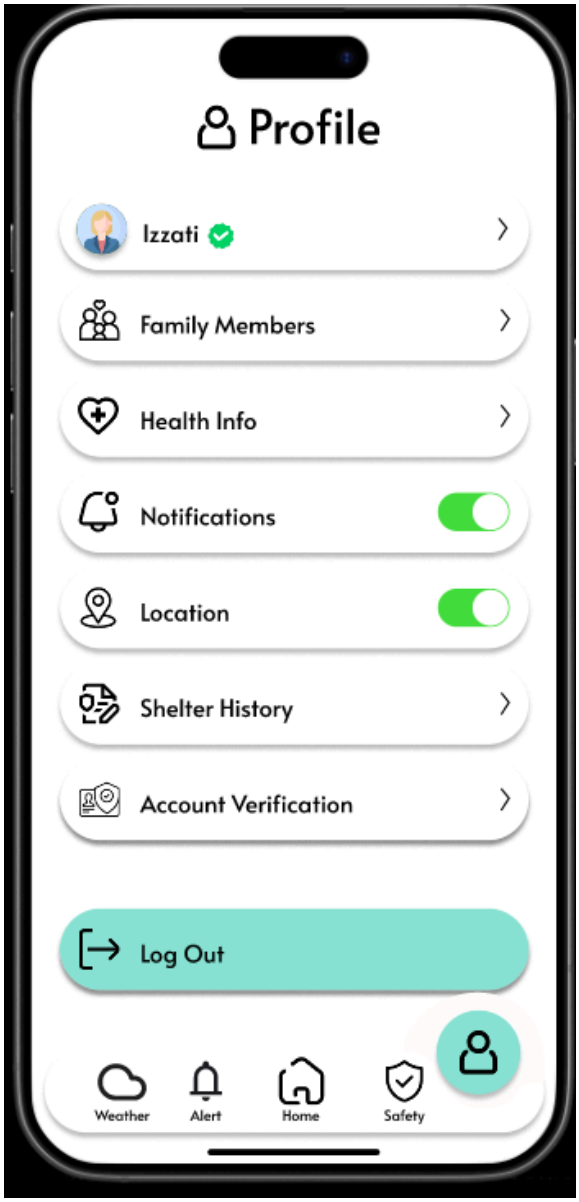


Figure 5.15 Profile

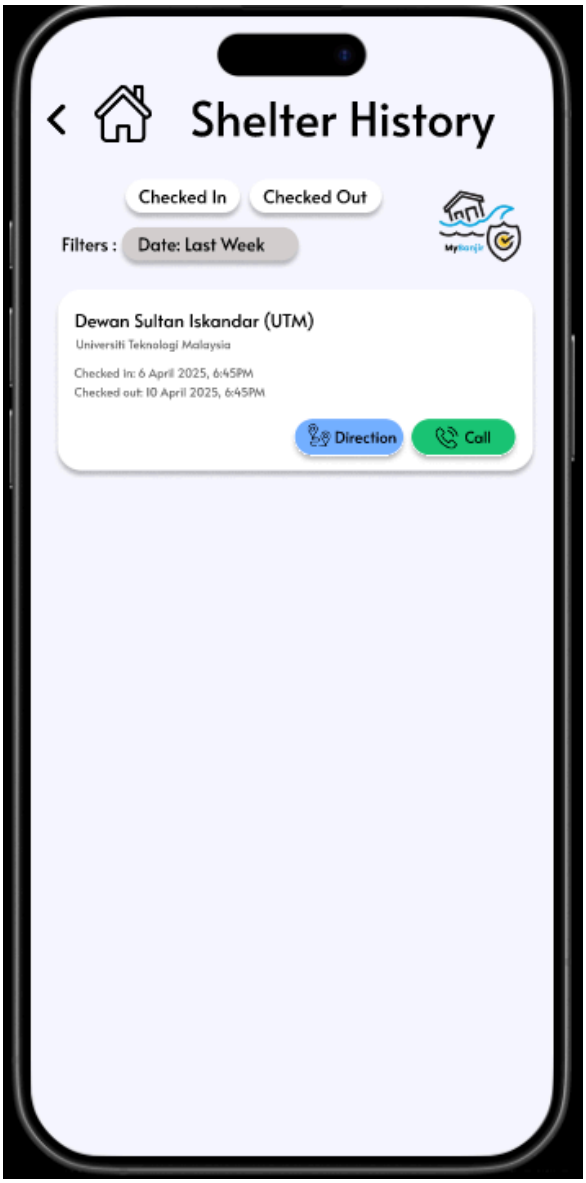


Figure 5.16 Shelter History

Figure 5.17 Account Verification

Figure 5.18 Face Verification

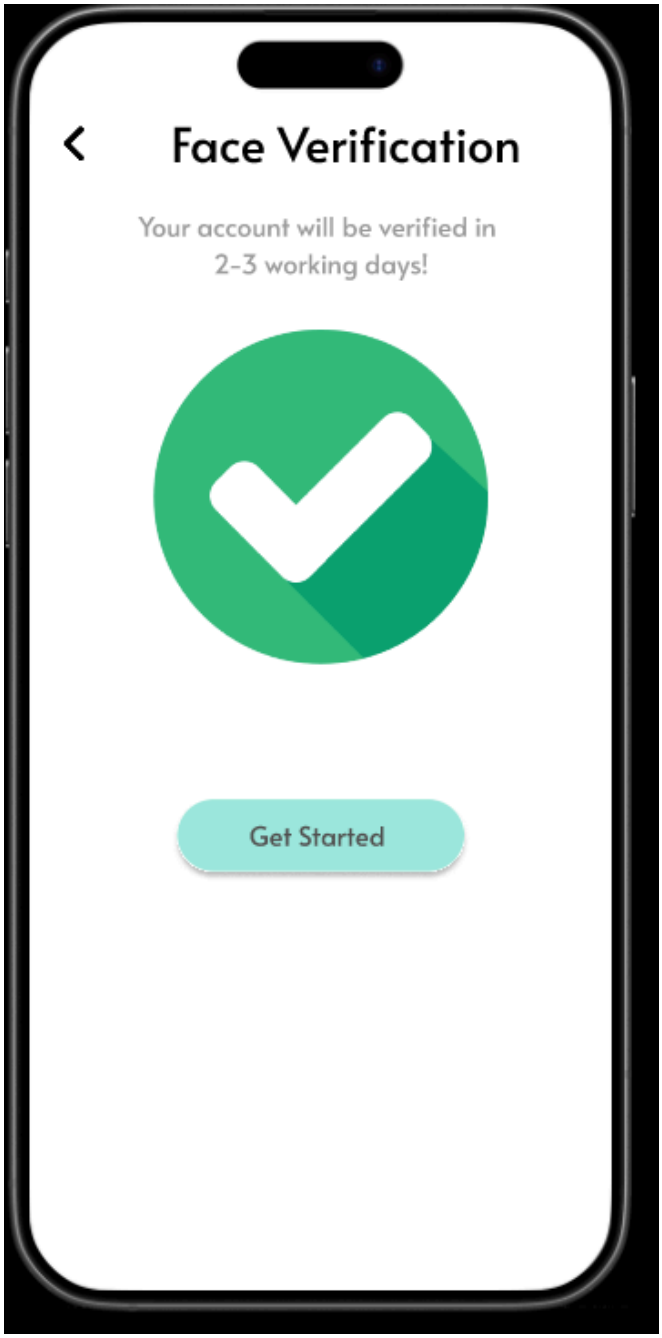


Figure 5.17 Successful Face Verification

6.0 Result

The development of the MyBanjir resulted in a functional prototype that integrates artificial intelligence with real-time environmental monitoring to enhance disaster preparedness and response. The system demonstrates how AI technologies can be applied to analyze environmental data, predict flood risks, and support decision-making during emergency situations. Several key AI-driven features were successfully implemented in the prototype, as described below.

6.1 AI-Based Weather Monitoring and Flood Prediction

One of the core components of the system is the AI-powered weather monitoring and flood prediction feature. This module collects environmental data such as rainfall intensity, river water levels, terrain elevation, and historical flood records from various external data sources.

Artificial intelligence models analyze these datasets to estimate the probability of flooding and predict the potential flood depth in specific locations. The system then presents this information to users through the Weather Page, allowing them to monitor real-time environmental conditions and understand potential risks.

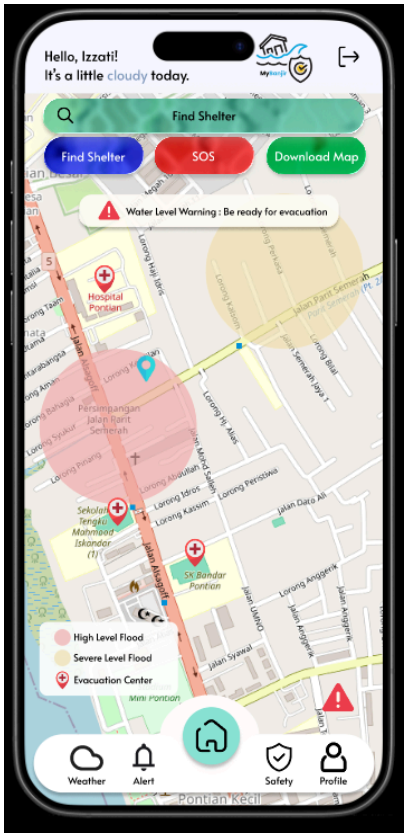
The AI prediction model can generate alerts indicating the likelihood of flooding within a short time frame, enabling users to take precautionary actions before the situation becomes critical. This predictive capability significantly improves early disaster preparedness compared to conventional flood monitoring systems.

6.2 AI-Assisted Safe Evacuation Route Guidance

The system also incorporates an AI-assisted evacuation route guidance feature that helps users identify safer paths during flood emergencies. Unlike traditional navigation systems that prioritize travel speed or shortest distance, the proposed system considers environmental risks when recommending evacuation routes.

The AI model evaluates predicted flood depth, potential road hazards, and geographical information to identify routes that minimize exposure to flooded or dangerous areas. By integrating this information with map navigation services, the system can guide users to safer locations such as evacuation shelters.

This intelligent routing system enhances the safety of evacuation processes and reduces the likelihood of individuals encountering hazardous conditions while attempting to escape flood-affected areas.



6.3 AI-Enhanced Community Disaster Monitoring

Another important result of the system is the implementation of community-based disaster monitoring supported by AI analysis. Users are able to report flood conditions, road blockages, or other hazards through the mobile application by submitting descriptions and images.

Artificial intelligence technologies can assist in verifying and interpreting the submitted information, enabling the system to update the disaster monitoring map more accurately. This crowdsourced data contributes to a more comprehensive understanding of the real-time disaster situation.

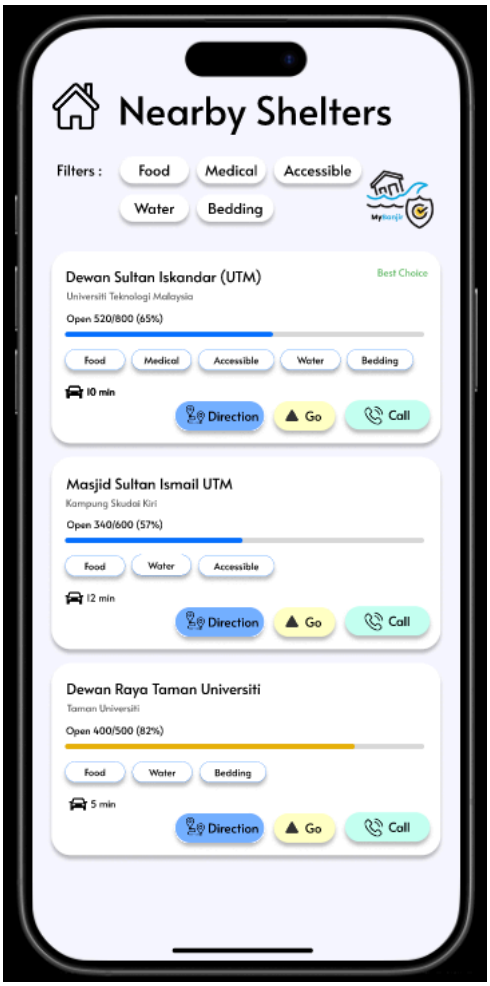
By combining official environmental data with community reports, the system improves situational awareness and enables faster response to emerging risks.

6.4 AI-Supported Shelter Recommendation and Resource Optimization

The system also demonstrates the use of AI-driven decision support for evacuation shelter management. Through integration with digital identity systems, users can quickly register at shelters and access relevant emergency services.

AI algorithms analyze user location, shelter capacity, and available resources to recommend suitable evacuation centers for affected individuals. This intelligent recommendation system helps distribute evacuees more efficiently and prevents overcrowding in certain shelters.

Such optimization improves disaster response coordination and ensures that available resources are utilized effectively during emergency situations.



6.5 Overall Impact of AI Integration

Overall, the results demonstrate that integrating artificial intelligence into the MyBanjir system significantly enhances disaster management capabilities. The AI-driven modules enable predictive flood monitoring, safer evacuation planning, improved situational awareness, and more efficient resource management during emergencies.

These outcomes support the project's objective of improving disaster preparedness and reducing the impact of flood events on communities. By providing timely information and intelligent decision-support tools, the system has the potential to strengthen resilience against climate-related disasters.

7.0 Conclusion

Flood disasters remain a critical challenge across Asia, particularly in regions affected by seasonal monsoons and extreme weather patterns. As climate change continues to intensify rainfall events and increase the frequency of natural disasters, the need for efficient disaster preparedness and early warning systems becomes increasingly important. Countries across Asia, including Malaysia, regularly experience flooding that disrupts communities, damages infrastructure, and threatens human lives. These recurring events highlight the urgent need for innovative technological solutions that can support public safety and disaster response.

The MyBanjir mobile application aims to address these challenges by providing an accessible and intelligent platform that enhances flood awareness and preparedness among communities. By delivering real-time flood alerts, safety information, and situational updates, the application helps users make informed decisions during emergency situations. Access to timely information can significantly reduce confusion and panic during disasters, allowing individuals and families to respond more quickly and safely.

A key feature of MyBanjir is the integration of Artificial Intelligence (AI) to improve disaster management capabilities. Through AI-powered analysis, the application can assist in predicting potential weather patterns and flood risks, enabling earlier warnings and better preparedness. Additionally, AI can help determine safer and more efficient evacuation centres by analysing factors such as flood-prone areas, accessibility, population density, and real-time environmental data. This intelligent approach supports authorities and communities in identifying evacuation locations that minimize risk and maximize safety during emergencies.

Furthermore, the use of predictive technologies aligns with global disaster risk reduction initiatives such as the Sendai Framework for Disaster Risk Reduction, which encourages the use of advanced technologies, data analysis, and early warning systems to reduce disaster-related losses. By leveraging mobile technology and AI-driven insights, MyBanjir contributes to building more resilient communities and improving disaster preparedness strategies across flood-prone regions in Asia.

In conclusion, the development of the MyBanjir application demonstrates the importance of combining technology, real-time data, and artificial intelligence to enhance flood safety and emergency response. The application not only provides vital information to the public but also supports smarter decision-making during disaster situations. As floods continue to pose a significant threat in Asia, solutions like MyBanjir play an essential role in protecting lives, improving evacuation planning, and strengthening community resilience against future disasters.